Problem Chosen	2026 HiMCM Summary Sheet	Team Control Number
A		18213

Choosing a City for High-School Life with Incomplete Evidence

Summary

Which of these sixteen cities best supports everyday life for a high schooler? We group ten everyday measures into five areas: safety and environment, cost of living, education, mobility and daily life. We use AHP weights from a high schooler's point of view, with safety and education at 34% each, and TOPSIS to give each city a score from 0 to 100.

Zurich ranks first at 64.5, with Madrid at 63.3 and Singapore at 62.5 close behind. Its strong safety record, higher-education options, transit and public pools put it ahead.

Zurich stays first in all 31 alternative ways of counting transit stops and pools and measuring travel. It also leads in about 83% of 10,000 random weightings around our chosen priorities (concentration 100) and has the smallest worst-case shortfall behind the best rival, 6.1 points. Madrid moves to first if Zurich's homicide figure, based on two offences in 2025, is treated as unknown. Madrid also moves to first if all post-2025 data are set aside.

Our three hypothetical 2035 scenarios keep the same top three. Families should shortlist Zurich, Madrid and Singapore, then check factors we could not measure, such as school eligibility, language and cultural fit.

Keywords: high-school city choice; AHP; TOPSIS; interval evidence; maximum regret; student priorities.

Contents

1 Introduction and decision scope 3
1.1 What makes a city suitable for high-school life? 3
1.2 Criterion selection and contribution 3
2 Assumptions and their role 3
3 Notation 4
4 Indicator construction and city selection 4
4.1 City selection 4
4.2 Indicator system 6
4.3 Data sources and filled values 10
5 Establishing the baseline evaluation model 12
5.1 Determining indicator weights (AHP) 12
5.2 TOPSIS scoring 13
5.3 Scoring results and interpretation 14
5.4 A significant revision: from equal weights to AHP weights 15
5.5 Robustness of the recommendation 15
6 Model revision and extension 18
6.1 Looking ahead to 2035 18
6.2 Within-city fairness and growth stages 19
7 Conclusion and decision use 21
News article 22
References 23
Report on Use of AI Tools (additional pages) 26

1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A student's school day depends on safe surroundings, affordable living, teaching resources and convenient travel. We compare sixteen cities, one per country, on ten everyday measures. Under our high-schooler priorities, Zurich leads Madrid by 1.2 points. We test how source choices, uncertain inputs and different priorities affect this recommendation in Section 5.5.

We express each measure as a utility from 0 to 1, with higher values preferred. The **representative score** is the TOPSIS score calculated at the midpoint of each cell's utility range.

1.2 Criterion selection and contribution

Our criteria cover safety, air quality, housing and living costs, teaching resources, higher education, transport stops, journey time, broadband and pools (Table 3). Friendships, language and cultural fit depend on the student, and comparable surveys of wellbeing and discrimination were unavailable. City totals also omit fees, eligibility and neighbourhood routes. Families should check these factors against their needs and budget.

We give pupils per teacher more AHP weight than pools because schooling shapes daily life. Compared with equal weights, twelve cities change rank; Zurich remains first. The appended AI-use report describes assistance with research, calculations and writing [7, 8].

2 Assumptions and their role

1. **Source scopes stay fixed.** We retain each value's year, boundary, population and school stage; comparisons describe those periods.

2. **Utilities use fixed scales.** Reference levels set the scales; more resources and lower burdens improve utility.

3. **Weights express our student priorities.** We assign safety and education 0.340 each, cost and mobility 0.123 each, and daily life 0.073, without a survey.

4. **Strengths can offset weaknesses.** Families should apply essential safety or eligibility thresholds before using the combined score.

5. **Utility ranges vary independently.** We vary the 37 utility ranges independently in endpoint tests; possible winners and maximum regret depend on it.

6. **Scenarios hold the comparison basis fixed.** We keep population denominators, reference levels and weights unchanged.

3 Notation

Symbol	Meaning
x, a	Observed indicator value and reference level (in the same units)
u, u_{ij}	Utility on $[0, 1]$; i identifies a city and j an indicator
c, P, d	Count, population and density (institutions per million; stops and pools per 100,000)
t, q	Travel duration in minutes and its utility
w_j	Indicator weight, with all ten weights summing to one
D_i^+, D_i^-	City i's distances to the ideal and worst utility profiles
S_i	City i's TOPSIS score, on $[0, 100]$
ℓ_{ij}, h_{ij}	Lower and upper utility bounds
L_i, U_i	Lower and upper score bounds, $S(\ell_i)$ and $S(h_i)$
R_i	Maximum regret
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Indicator construction and city selection

4.1 City selection

We compare sixteen cities: Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. The set covers sixteen countries and a range of costs, climates, population sizes, mobility and safety (Table 2). Fourteen cities are European and all are in high-income economies, so our recommendation applies to these sixteen candidates. Figure 1 shows their locations and final ranks.

We choose one city per country to avoid repeating national data systems and school structures. We prioritise city records, using national statistics or surveys as labelled proxies (Section 4.3). Table 1 counts the evidence labels. Withdrawal tests examine team-table values without original links in Section 5.5.

Table 1: City profiles and evidence labels. Direct values are official or team-table values; ranges and proxies are counted separately.

City	Country	Population (year)	Direct values	Utility ranges	Proxies
Vienna	Austria	2,028,399 (2025)	8	1	1
Zurich	Switzerland	440,935 (2025)	6	3	1
Helsinki	Finland	694,392 (2025)	7	2	1
Berlin	Germany	3,700,577 (2025)	8	2	0
London	United Kingdom	9,122,909 (2025)	7	2	1
Amsterdam	Netherlands	941,927 (2025)	7	2	1
Stockholm	Sweden	999,200 (2025)	6	4	0
Paris	France	2,103,778 (2023)	7	2	1
Madrid	Spain	3,527,924 (2025)	7	2	1
Milan	Italy	1,399,079 (2025)	6	3	1
Dublin	Ireland	592,713 (2022)	7	2	1
Oslo	Norway	727,797 (2025)	8	2	0
Prague	Czechia	1,407,084 (2025)	7	3	0
Warsaw	Poland	1,863,800 (2024)	7	2	1
Singapore	Singapore	6,036,900 (2024)	7	3	0
Tokyo	Japan	9,983,365 (2026)	7	2	1
Total			112	37	11

Table 2: The sixteen cities distribute across the dimensions the indicators measure.

Dimension	Span within the set
Cost of living	Moderate capitals (Warsaw, Prague, Madrid, Berlin) to among the world's most expensive (Zurich, Oslo, London, Singapore)
Climate	Cold northern (Oslo, Helsinki, Stockholm) and continental (Berlin, Vienna, Warsaw, Prague) through maritime (London, Dublin, Amsterdam) and warm Mediterranean (Madrid, Milan) to tropical Singapore and humid-subtropical Tokyo
Scale	Mid-size capitals (Zurich, Oslo, Helsinki, Dublin) up to large metropolises (Tokyo, London, Paris)
Mobility	Cycling and tram networks (Amsterdam, Berlin, Prague, Vienna) to dense metro systems (Tokyo, Paris, London)
Safety	Low recorded homicide rates (Singapore, Tokyo, Oslo) alongside higher rates elsewhere in the set

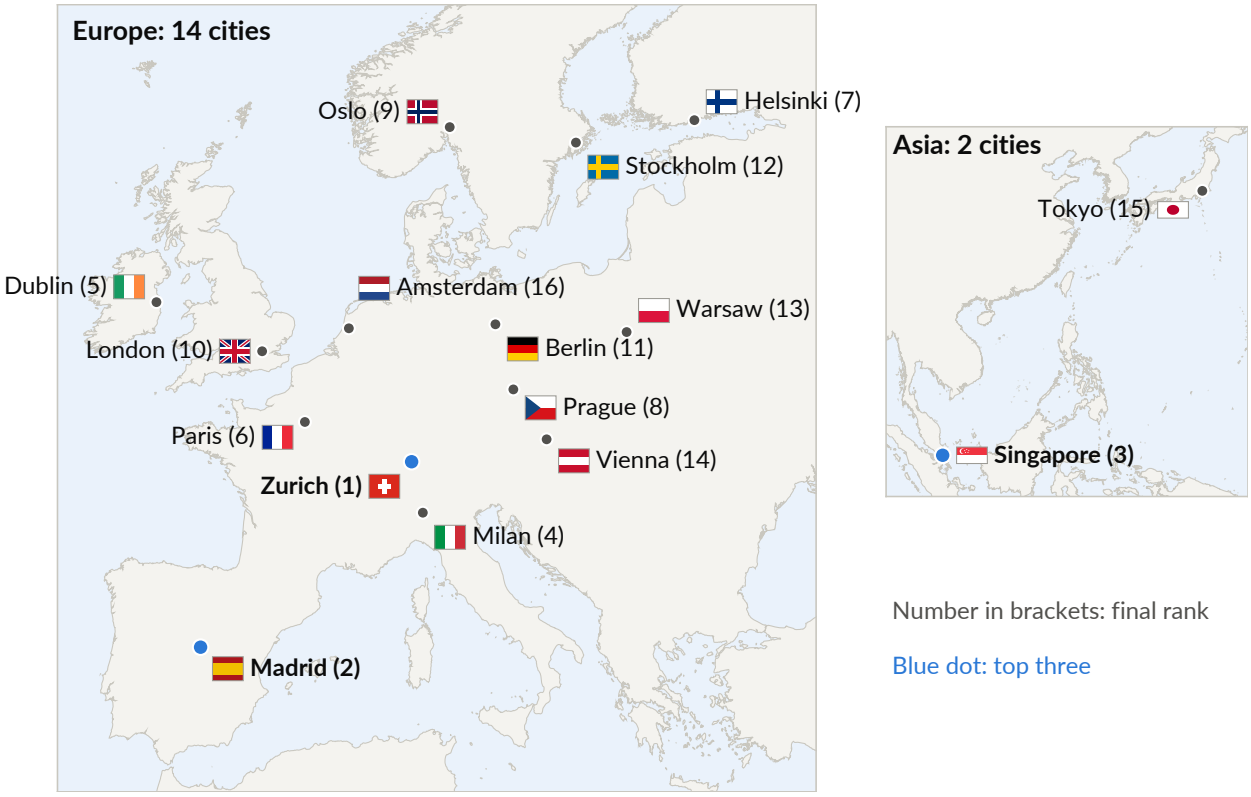


Figure 1: The 16 candidate cities, with national flags and final ranks in brackets; blue dots mark the top three.

4.2 Indicator system

We group the ten indicators into five categories, with two indicators in each. Table 3 gives their units, preferred directions, anchors and data notes. Higher utilities on $[0, 1]$ are always preferred.

We reward larger values for benefit indicators using

$$u = \frac{x}{a + x}, \tag{1}$$

and smaller values for cost indicators using

$$u = \frac{a}{a + x}. \tag{2}$$

The anchor a gives utility 0.5 when $x = a$. Higher-education density is $10^6 c/P$ institutions per million, with anchor 10 per million.

We compare transport stops and pools by density per 100,000 residents:

$$d = \frac{100000c}{P}. \tag{3}$$

We use $x = d$ in Equation 1. When other providers are uncounted, the verified subset sets the lower utility bound and the upper bound remains 1.

Table 3: Indicator definitions, anchors and data treatment.

Category	Indicator	Unit	Better when	Anchor a	Data note
Safety and environment	Homicide rate	Deaths per 100,000	Lower	2 deaths per 100,000	City record; Zurich recalculated
	$PM_{2.5}$ concentration	$\mu g/m^3$	Lower	15 $\mu g/m^3$	City record
Cost of living	Living-cost index	Index points	Lower	100 points	City record
	Housing price-to-income ratio	Ratio	Lower	10	City record
Education	Upper-secondary pupils per teacher	Pupils per teacher FTE	Lower	15 pupils/FTE	City record; proxy fills
	Higher-education institutions	Institutions per million	Higher	10 per million	City record
Mobility	Mean travel/commute time	Minutes	Lower	30-minute decay scale	City record; survey proxies and utility ranges
	Public-transport stops	Stops per 100,000	Higher	50 per 100,000	City record; utility ranges for uncounted providers
Daily life	Fixed broadband speed	Mbps	Higher	100 Mbps	City record
	Public swimming pools	Venues per 100,000	Higher	1 per 100,000	City record; utility ranges for uncounted providers

We measure upper-secondary teaching resources as pupils per full-time-equivalent (FTE) teacher:

$$x = \frac{\text{upper-secondary pupils}}{\text{upper-secondary teacher FTE}}. \tag{4}$$

Prague's 2024/25 records give about 11.6 pupils per teacher after lower-gymnasium years are removed [5].

Shorter journeys receive higher utility. For a journey of t minutes, we use

$$q = \exp(-t/30). \tag{5}$$

When we have only a mean duration $\bar{t}$, we use the utility range

$$\mathbb{E}[q] \in [\exp(-\bar{t}/30), 1]. \tag{6}$$

Convexity gives the lower endpoint; 1 is the upper limit for nonnegative durations. We use work-commute means to describe general travel conditions and Singapore's school-stage bins to form a separate utility range [12, 6].

The cities differ across indicators (Figure 3). Singapore and Milan have the highest homicide utilities, 0.97 and 0.89, while Amsterdam has 0.06. Madrid has the lowest pupil-teacher proxy ratio, 9.6 pupils per teacher, and utility 0.61. Figure 2 shows the raw values, including Zurich's higher-education supply of 52 institutions per million.

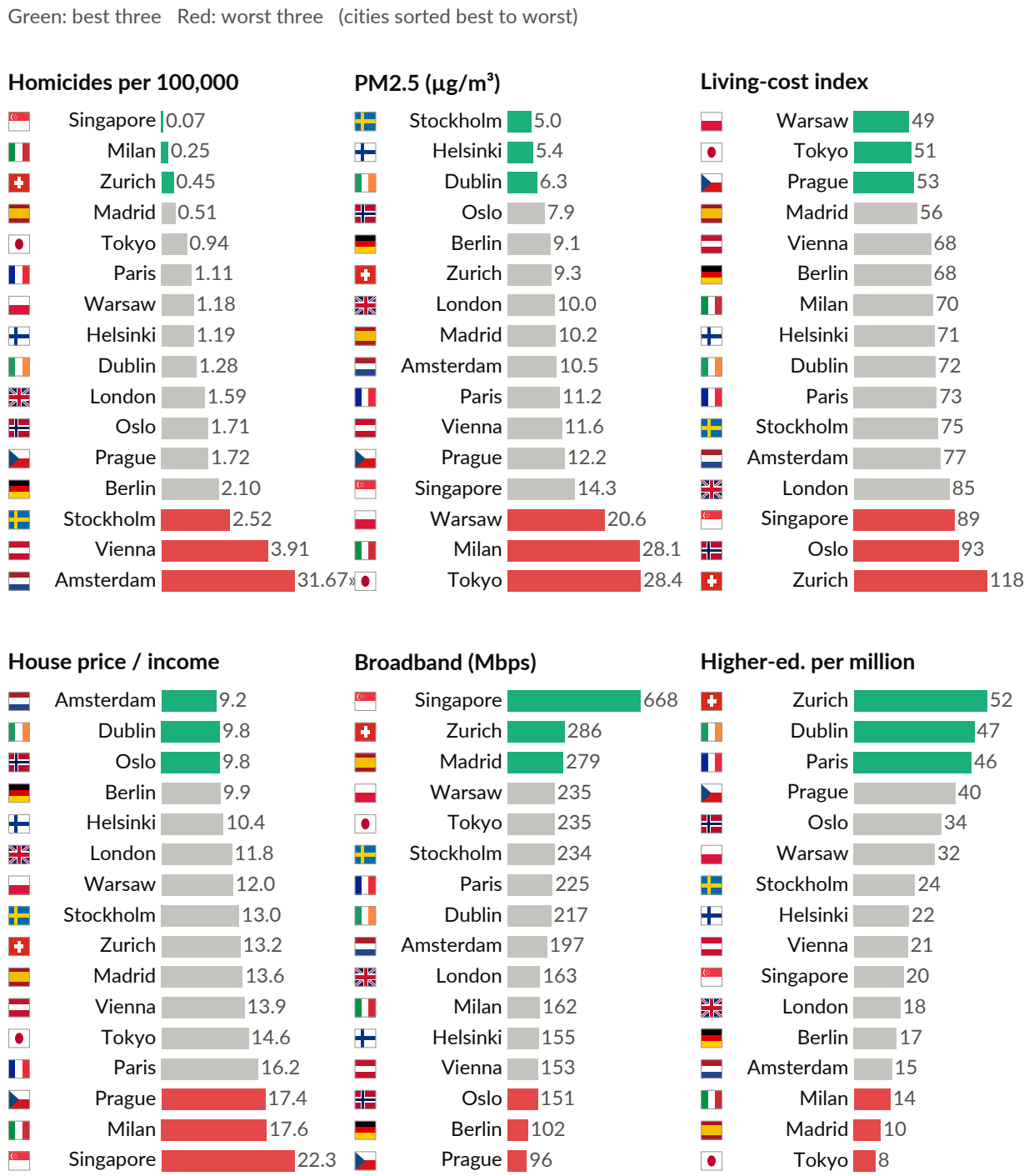


Figure 2: Raw values for the six indicators with complete city records, ordered from best to worst within each indicator. The best three cities are green and the worst three red; flags identify countries. Homicides are shown per 100,000, with Zurich's recalculated 0.45 and Amsterdam's 31.67 marked off-scale with "≫". Higher-education supply is institutions per million.

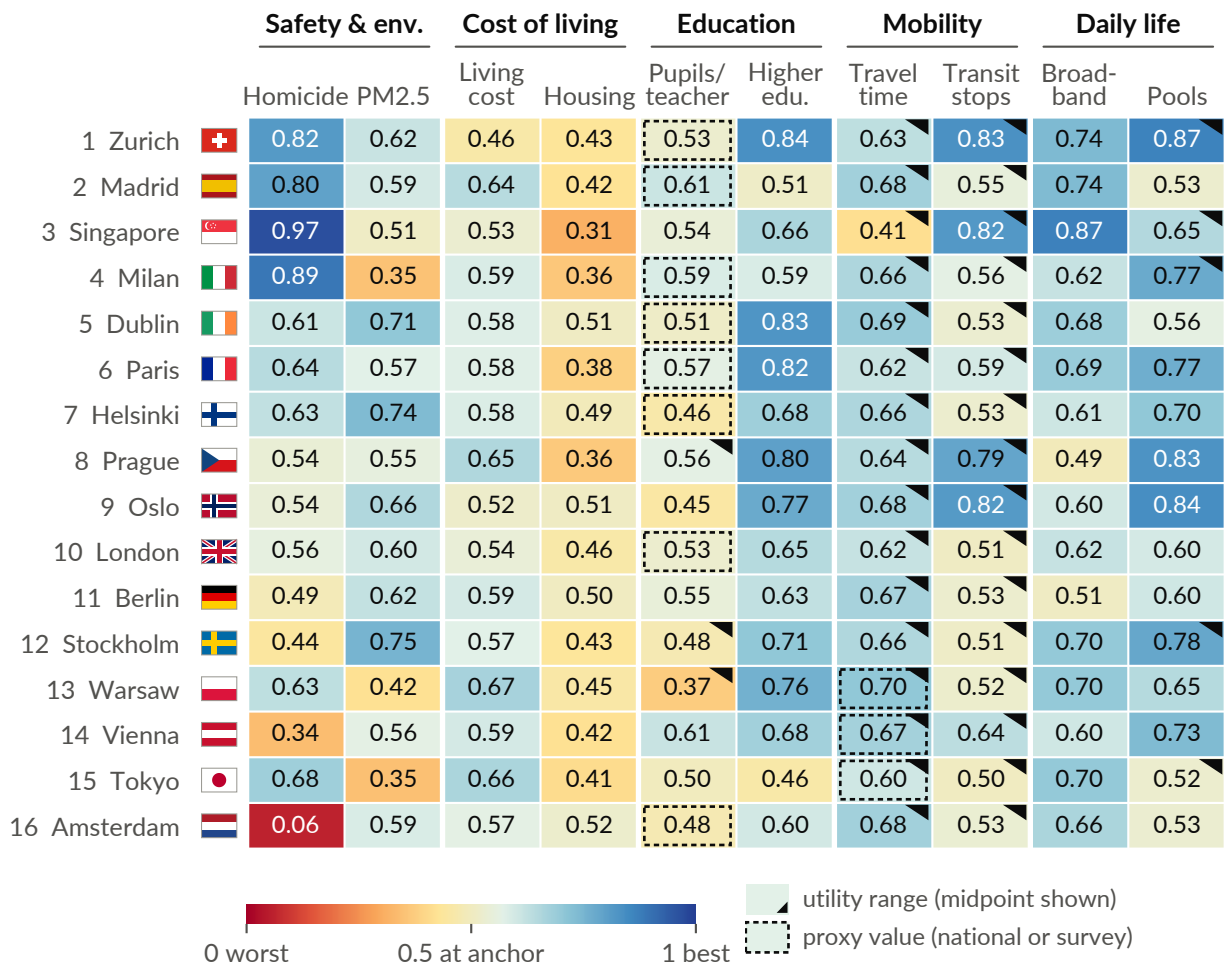


Figure 3: Indicator utilities for the sixteen cities, ordered by final score and grouped by category. Colour is utility from 0 (worst) to 1 (best). Utility ranges show their midpoint with a black corner triangle, proxies have a dashed outline, and flags mark countries.

4.3 Data sources and filled values

We build the comparison from the team table, retaining its source years and boundaries [1]. Table 4 lists the main sources. We apply three rules to keep measurements comparable:

1. Count each station once, excluding separate platforms and repeated line entries. Warsaw's 21 M1 and 18 M2 entries give 38 stations after merging the shared Świętokrzyska interchange [14].
2. Match school stage before dividing pupils by teacher FTE. Prague's totals therefore remove lower-gymnasium pupils and teachers [5].
3. Recalculate Zurich's supplied 0.14 as 0.45 per 100,000: two completed homicide offences recorded by the Zurich city police in 2025, divided by 440,935 residents [11]. A rate built on two events is unstable.

We count 38 distinct stations in Tokyo's Oedo directory within the 23 wards. Oslo's inventory uses parent stop groups; we exclude the questioned Tveten bru entry because its closure is unconfirmed [13, 4].

We fill gaps with 11 proxy cells. Six are Eurostat national pupil-teacher ratios from table educ_uoe_perp04 (ISCED 3, 2024): Helsinki 17.4, Amsterdam 16.4, Dublin 14.2, Paris 11.5, Milan 10.6 and Madrid 9.6. The two median pupil-teacher proxies, for Zurich and London, use 13.45. Three survey travel means enter as utility ranges: Vienna 32 minutes, Tokyo 47.5 minutes from the 2021 Japanese time-use survey, and Warsaw 27 minutes [9].

Our matrix contains 160 entries: 123 single values, including proxies, and 37 utility ranges. Zurich's homicide recalculation is separate from the 11 proxies.

Table 4: Sources and years by indicator. Supplements establish counted subsets, proxies and boundary rules for the team matrix.

Indicator	Main sources	Year(s)	Notes
Homicide rate	Team table; Zurich police statistics [1, 11]; CBS recount for Amsterdam [10]	2022–2025	Zurich: two completed offences recorded by city police in 2025; 440,935 residents; rate 0.45 per 100,000.
$PM_{2.5}$	Team table	2021–2025	City monitoring values with reported station and year scopes.
Living-cost index	Team table [1]	2025–2026	City index values; source year stays attached to each row.
Housing price-to-income	Team table; Oslo property-price archive [1, 58]	2025–2026	City price-to-income values; administrative boundaries follow the team matrix.
Upper-secondary pupils per teacher	Team table; Prague, Paris, Finland and Ireland education records; Eurostat proxy fills [1, 5, 24, 54, 33, 53, 48, 23, 26, 27]	2024–2026	Six Eurostat national ratios and two median fills for Zurich and London are labelled proxies; Prague is stage-matched; Shanghai and Singapore stage tables support supplementary records.
Higher-education institutions	Team table; school and post-secondary directories [1, 40, 41]	2025–2026	Institutions per million; each provider counted once, with cross-stage providers flagged.
Mean travel/-commute time	Team table; Eurostat city series; Warsaw school-trip survey; Singapore census bins [1, 12, 9, 6, 25]	2011–2024	Vienna, Tokyo and Warsaw means enter as utility ranges; work-commute means are proxies for general travel conditions.
Public-transport stops	Operator and municipal inventories for Oslo, Zurich, Helsinki, Madrid, Vienna, Prague, Berlin, London, Singapore, Amsterdam, Milan, Stockholm, Tokyo and Warsaw [18, 4, 19, 20, 29, 34, 35, 36, 37, 38, 39, 42, 44, 55, 52, 51, 13, 14, 57, 30, 43, 56]	2024–2026	Operator inventories, counted once per station; interchange and boundary rules are explicit, with verified subsets forming lower bounds.
Fixed broadband speed	Team table	2025–2026	City speed observations with the reported test window.
Public swimming pools	Team table; municipal and operator inventories for Stockholm, Paris, Tokyo, Singapore, Zurich and Milan [1, 21, 22, 31, 32, 45, 46, 47, 49, 50]	2025–2026	Venues counted once per provider; operator inventories are lower bounds when coverage is incomplete.

5 Establishing the baseline evaluation model

We turn the city data into a ranking through the steps in Figure 4.

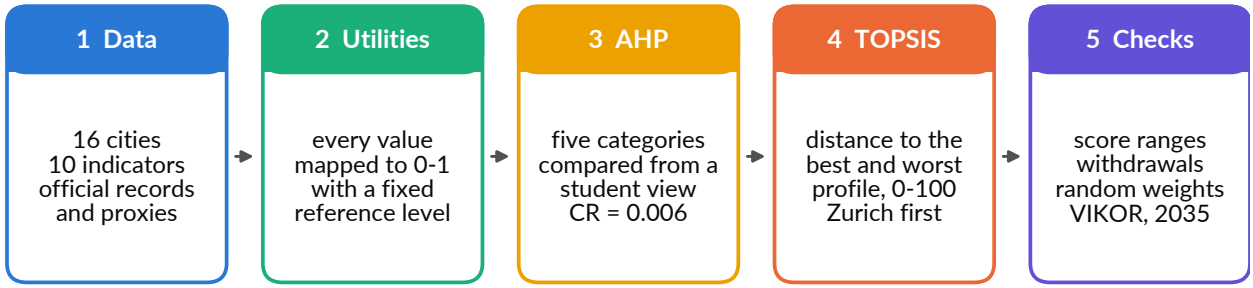


Figure 4: Model workflow: raw city data become utilities, AHP sets the weights, TOPSIS gives the representative score and ranking, and robustness checks test the result.

5.1 Determining indicator weights (AHP)

Step 1: Set up the hierarchy. We use the analytic hierarchy process (AHP) to weight five categories and ten indicators for our city choice [2].

Step 2: Compare the five categories. We prioritise safety and schooling, followed by cost and mobility, then broadband and pools. Table 5 uses Saaty's 1–9 scale: 1 means equal importance; reverse comparisons use reciprocals.

Table 5: Pairwise category comparisons from our student viewpoint. S: safety and environment; C: cost of living; E: education; M: mobility; D: daily life.

Category	S	C	E	M	D
Safety and environment	1	3	1	3	4
Cost of living	1/3	1	1/3	1	2
Education	1	3	1	3	4
Mobility	1/3	1	1/3	1	2
Daily life	1/4	1/2	1/4	1/2	1

Step 3: Calculate category weights and check consistency. We normalise the positive eigenvector v for matrix A's largest eigenvalue to sum to one, giving category weights g_k:

$$Av = \lambda_{\max} v. \tag{7}$$

For $n = 5$ categories, we calculate the consistency index CI and ratio CR:

$$CI = \frac{\lambda_{\max} - n}{n - 1}. \tag{8}$$

$$CR = \frac{CI}{RI}. \tag{9}$$

Using random index $RI = 1.12$, we obtain $\lambda_{\max} = 5.0264$, $CI = 0.00660$ and $CR = 0.0059 < 0.1$, meeting the consistency threshold [2].

Step 4: Compare the two indicators within each category. We use local importance ratios of 2:1 for homicide to $PM_{2.5}$, 1:1 for living costs to housing, 3:1 for teacher resources to higher-education supply, 2:1 for travel time to stops, and 2:1 for broadband to pools. Each pair gives a consistent matrix. We convert its ratio to local weights p_j, then global weights

$$w_j = g_k p_j. \tag{10}$$

Figure 5 shows both levels. We calculate with unrounded weights.

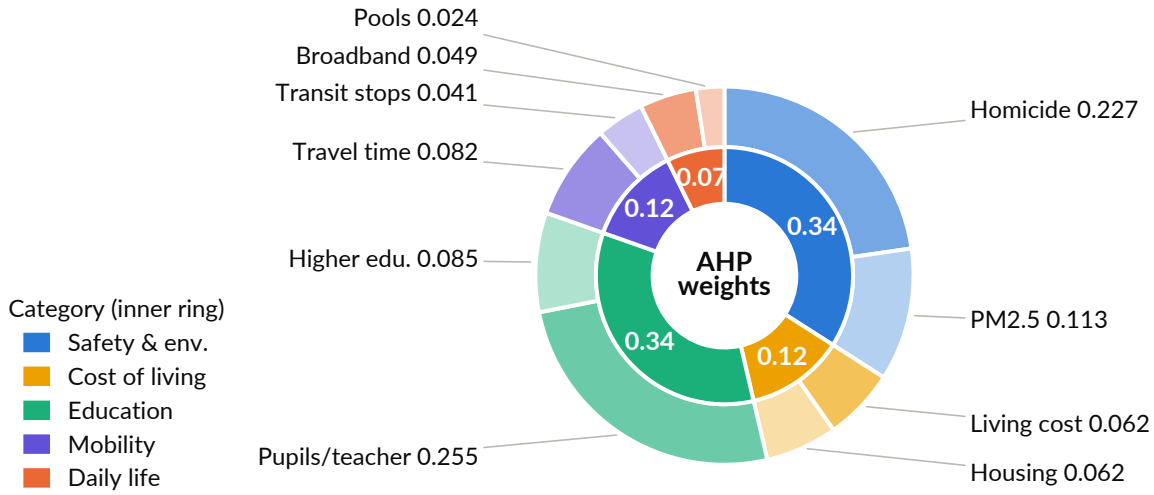


Figure 5: AHP weights from our student priorities. The inner ring shows the five category weights; the outer ring shows the ten global indicator weights, printed beside each slice. Safety and environment and education each receive 0.340.

5.2 TOPSIS scoring

Step 1: Form the utility matrix. We use each utility range's midpoint on the shared 0–1 scale for the representative score:

$$u_{ij} = \frac{\ell_{ij} + h_{ij}}{2}. \tag{11}$$

Step 2: Find the distance to the ideal. We measure weighted shortfalls from the all-one ideal; smaller D_i^+ is better [3]:

$$D_i^+ = \sqrt{\sum_{j=1}^{10} w_j (1 - u_{ij})^2}. \tag{12}$$

Step 3: Find the distance to the worst profile. We measure progress above the all-zero profile; larger D_i^- is better:

$$D_i^- = \sqrt{\sum_{j=1}^{10} w_j u_{ij}^2}. \tag{13}$$

Step 4: Calculate the score. TOPSIS combines these distances into representative score S_i, from 0 to 100:

$$S_i = 100 \frac{D_i^-}{D_i^+ + D_i^-}. \tag{14}$$

Step 5: Carry utility ranges into the score. Higher utilities raise the score, so we use the lower and upper vectors ℓ_i and h_i to bound it:

$$[L_i, U_i] = [S(\ell_i), S(h_i)]. \tag{15}$$

5.3 Scoring results and interpretation

Zurich leads with a representative score of 64.5, followed by Madrid and Singapore (Table 6). Figure 6 compares ranks 1–3, 7–9 and 14–16 as examples from each tier.

Table 6: Representative scores under AHP weights, using the midpoint of each utility range.

Rank	City	Score	Tier	Rank	City	Score	Tier
1	**Zurich**	**64.5**	**Top**	**9**	**Oslo**	57.6	Middle
2	**Madrid**	63.3	Top	**10**	**London**	56.1	Bottom
3	**Singapore**	62.5	Top	**11**	**Berlin**	55.6	Bottom
4	**Milan**	61.1	Middle	**12**	**Stockholm**	55.4	Bottom
5	**Dublin**	60.8	Middle	**13**	**Warsaw**	54.5	Bottom
6	**Paris**	60.8	Middle	**14**	**Vienna**	54.4	Bottom
7	**Helsinki**	58.7	Middle	**15**	**Tokyo**	54.0	Bottom
8	**Prague**	58.4	Middle	**16**	**Amsterdam**	45.2	Bottom

Top tier. Ranks 1–3 score 62.5–64.5, led by safety and environment and education. These category scores are 0.75 and 0.61 for Zurich, 0.73 and 0.59 for Madrid, and 0.81 and 0.57 for Singapore.

Middle tier. Ranks 4–9, Milan through Oslo, score 57.6–61.1, starting 1.4 points below Singapore. Their strengths differ: Helsinki scores 0.66 for safety and environment and 0.52 for education; Prague scores 0.62 for education and 0.54 for safety and environment.

Bottom tier. Ranks 10–16, London through Amsterdam, score 45.2–56.1, starting 1.4 points below Oslo. Low utilities in heavily weighted indicators pull scores down: Vienna scores 0.41 for safety and environment and 0.63 for education; Amsterdam's homicide utility of 0.06 leaves it 8.8 points behind Tokyo.

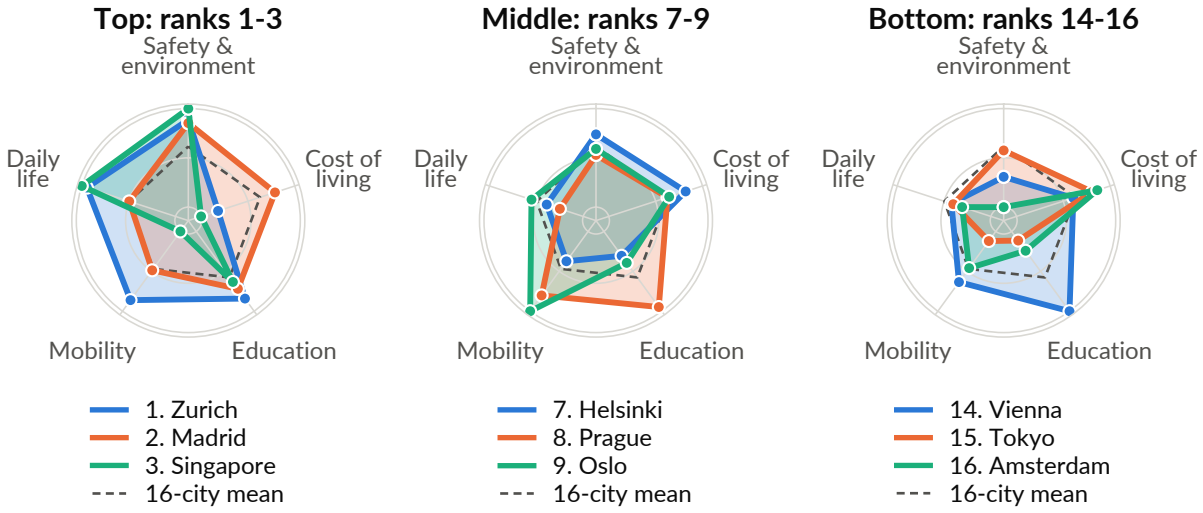


Figure 6: Representative cities chosen to contrast tiers: the whole top tier (ranks 1–3), the lower end of the middle tier (7–9), and the lowest three cities (14–16). Each axis is stretched from the lowest city (centre) to the highest city (outer ring) among all sixteen; the dashed line is the sixteen-city mean. Category scores combine each pair of indicators using its local AHP weights.

5.4 A significant revision: from equal weights to AHP weights

Our revision gives safety and schooling more influence. Equal weights assign every indicator 0.1; under AHP, safety and environment and education each receive 0.340, cost and mobility 0.123 each, and daily life 0.073.

With utilities, anchors and scoring unchanged, the leading four shift from Zurich, Oslo, Paris and Dublin to Zurich, Madrid, Singapore and Milan. Twelve of sixteen cities change rank; Kendall's $\tau = 0.517$ measures their agreement. Zurich stays first.

5.5 Robustness of the recommendation

Score bounds and maximum regret. We test utility ranges and weights in Figure 7. Maximum regret is the most a city could trail its best rival when every utility range works against it:

$$R_i = \max\left\{0, \max_{k \neq i} U_k - L_i\right\}. \tag{16}$$

Zurich has the smallest maximum regret, 6.1 points; nine cities can be first under some endpoint combination.

Dependence on homicide inputs. Zurich stays first up to about 0.69 per 100,000 (about three offences a year), where Madrid draws level. Withdrawing Zurich's homicide value, using utility range $[0, 1]$ for an unknown, puts Zurich eighth and Madrid first. Official

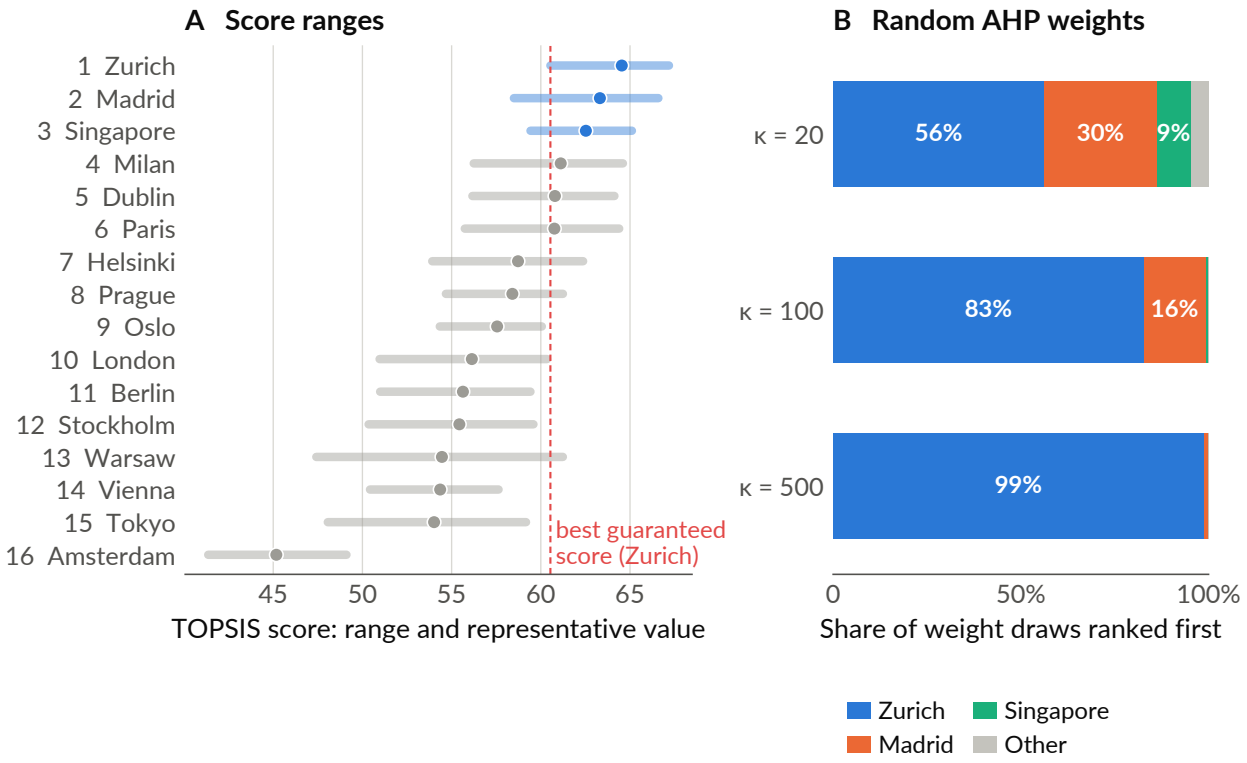


Figure 7: Robustness of the recommendation. Panel A shows each city's score interval as a light bar and its representative score as a dot; the top three are blue and the dashed line is Zurich's lower bound. Panel B shows the share of 10,000 AHP-centred Dirichlet weight draws in which Zurich, Madrid, Singapore or another city ranks first at concentrations 20, 100 and 500.

CBS counts do not reproduce Amsterdam's supplied numerator; the same withdrawal test lifts it to fifteenth [10].

Withdrawal tests. We replace selected point cells with utility range $[0, 1]$; Table 7 shows the resulting winners.

Table 7: First city after withdrawing dated or flagged point cells.

Withdrawn group	Cells	First (AHP)	First (equal weights)
Before 2020	13	Zurich	Zurich
After 2025	64	Madrid	Zurich
Flagged entries	28	Milan	Zurich
Union of the three groups	95	Milan	Milan

Source choices. Zurich leads under AHP and equal weights in the baseline, and under AHP in all 31 source-variant checks (Table 8). These change counting rules and sources; Table 7 gives broader withdrawal tests.

Table 8: The 31 source-variant cases in nine groups, including the before-update and current comparison cases.

Group	Cases	What changes	First city
Before-update baseline	1	Baseline before the source refinements	Zurich
Current comparison	1	Conditional source refinements included	Zurich
Prague bus inventory	2	Withdraw refinement; withhold request-stop witnesses	Zurich
Vienna bus inventory	11	Withdraw refinement; withhold selected route witnesses in nine single-route cases and one nine-route case	Zurich
Oslo bus inventory	5	Withdraw refinement; withhold two rail-replacement stops, Midtstuen, or all three service flags; include questioned Tveten bru stop	Zurich
Singapore pools	5	Withdraw OCBC, directory refinement, or all supplements; withhold three or four notice venues	Zurich
Zurich pools	1	Withdraw refinement	Zurich
Metro inventories	4	Withdraw Tokyo, Warsaw, both, or Amsterdam mapped inventory	Zurich
Travel	1	Withdraw Singapore historical school-stage proxy	Zurich
Total	31		Zurich

Priority checks. We vary priorities in 10,000 draws, $w \sim \mathrm{Dirichlet}(\kappa w_{\mathrm{AHP}})$. Zurich leads in 56.3%, 82.8% and 98.7% at concentrations $\kappa = 20, 100, 500$; larger κ keeps weights closer to AHP. Zurich also stays first when we double education or mobility weights separately.

Compromise ranking. VIKOR balances total and largest shortfalls using the same utilities and weights. We scale shortfalls between the best and worst observed utilities, then

equally combine the normalized weighted total and maximum [17]. Madrid, Zurich and Paris lead. Madrid meets the stability condition; its lead is below the required $1/(16-1)$ advantage, leaving Madrid and Zurich as close alternatives.

6 Model revision and extension

6.1 Looking ahead to 2035

We test more public services (S1), higher costs and congestion (S2), and smaller classes, cleaner air and faster travel (S3) as possible 2035 scenarios. S1 increases higher-education institutions, transit stops and pools by 10%; S2 increases living costs, housing price-to-income and travel time by 10%; S3 reduces pupils per teacher, $PM_{2.5}$ and travel time by 10%. These hypothetical changes use the same formulas, anchors and weights in every city.

Figure 8 shows the top six: Zurich, Madrid and Singapore stay first, second and third, and Dublin and Paris swap fifth and sixth under S3. Ranks 7–16 do not change in any scenario. More services add 0.1–0.3 points, higher costs and congestion subtract 0.4–0.5, and smaller classes with cleaner air and faster travel add about 1.0.

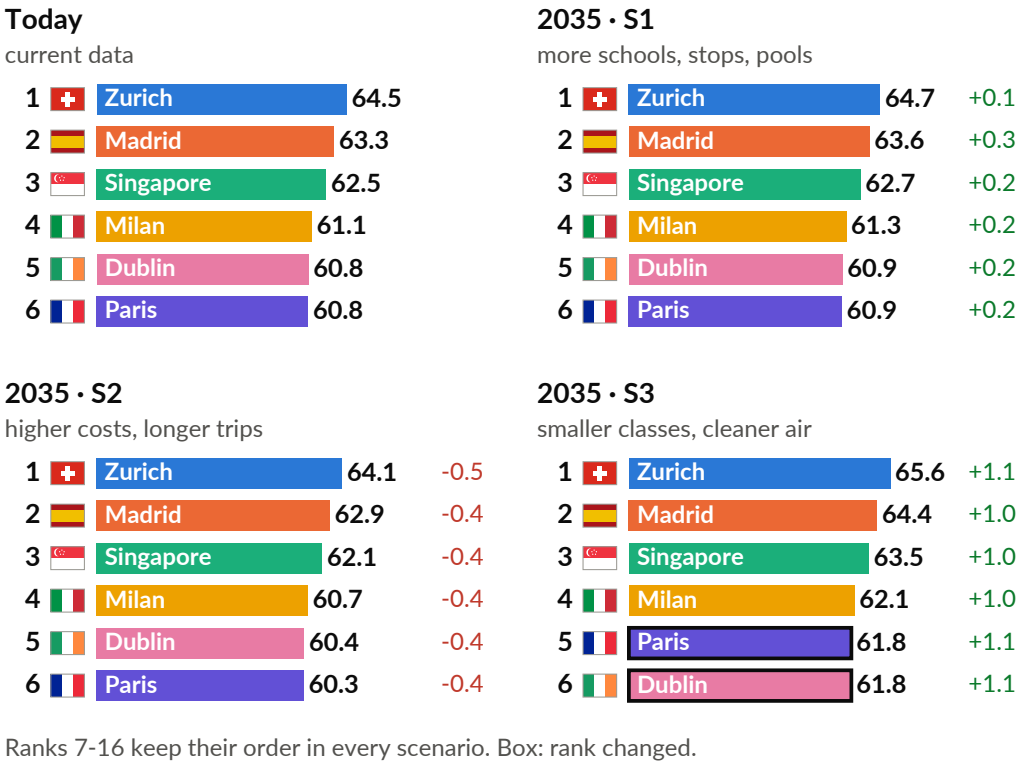


Figure 8: Top six cities today and under three hypothetical 2035 scenarios. Bars show representative scores; numbers give changes from today, and boxes mark changed ranks. S1: 10% more higher-education institutions, transit stops and pools. S2: 10% higher living costs, housing price-to-income and travel time. S3: 10% fewer pupils per teacher, lower $PM_{2.5}$ and shorter travel time.

6.2 Within-city fairness and growth stages

Citywide supply can mask differences in access and age. Figure 9 shows walking-time context from resident grids [15]. Istanbul's 2012 cohort (ages 14–18) has a minimum travel-utility gap of 0.227 between all students and the lowest fifth [28]. In Oslo's 2023 survey, 66% of grade 8 and 41% of final-year upper-secondary respondents rated youth meeting places positively [16].

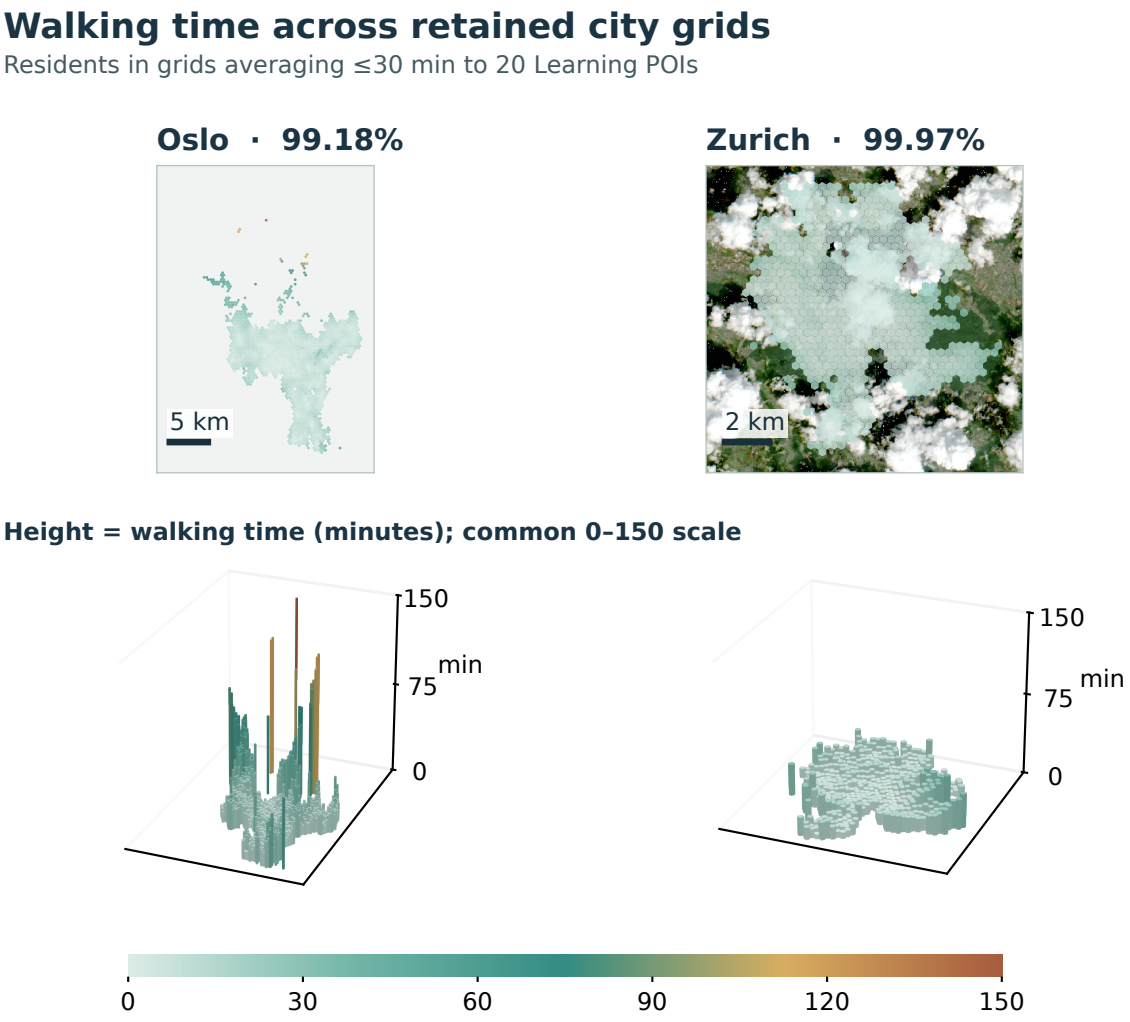


Figure 9: Walking-time context from the brunogrids resident grids. The modelled mean walking time is measured from resident grid cells to 20 learning POIs, with one common 0–150 minute scale. Among retained residents, 99.18% of Oslo and 99.97% of Zurich fall within 30 minutes. This context uses the source's resident grids and is not part of the ten-indicator score.

7 Conclusion and decision use

We recommend Zurich among these sixteen cities for its safety and education scores under our team's high-schooler priorities. It leads with AHP and equal weights, in all 31 source-variant checks, and in 82.8% of 10,000 weight trials at $\kappa = 100$. Its maximum regret is the smallest, at 6.1 points. Madrid trails by 1.2 points and leads when we withdraw Zurich's homicide value, treating it as unknown with utility range $[0, 1]$. Broader withdrawal tests of dated or flagged entries can put Madrid or Milan first (Table 7). All three hypothetical 2035 scenarios keep Zurich, Madrid and Singapore in the top three. Families should use this shortlist to compare schools, admission, fees and daily routes with their student's needs.

A city choice built around a student's day

Picture the journey from the front door to the classroom, then to a swim after school. Which city would suit that day? We recommend Zurich among sixteen cities, with 64.5 points. Madrid follows just 1.2 points behind; Singapore takes third.

What matters during a school day?

We built our comparison around ten everyday measures: safety, air quality, living and housing costs, teaching resources, higher education, transport stops, travel time, broadband and swimming facilities. Our team gave safety and education the greatest influence. Zurich's school resources, safety, transport and daily facilities help balance its higher costs.

Zurich stays first when we change counting rules and sources for stops, pools and travel in 31 checks. Treating all 64 figures dated after 2025 as unknown puts Madrid first. Doing the same for 28 figures with data concerns, or all 95 covered by either check, puts Milan first. Zurich also leads in 82.8% of 10,000 alternative priority choices close to ours. When uncertain evidence counts against each city, Zurich falls fewer points behind its best rival than any other choice.

Why Madrid deserves a close look

Madrid is close enough for a small change in safety data to tip the choice. Zurich's homicide rate is 0.45 per 100,000, based on two completed homicide offences recorded by city police in 2025 and 440,935 residents [11]. With so few events, the rate can shift sharply. Zurich stays first up to about 0.69 per 100,000, about three offences a year. Treating that figure as unknown puts Madrid first.

Eleven figures help fill gaps in school and travel data: six national pupil-teacher ratios from Eurostat, two median ratios for Zurich and London, and three average travel times from surveys, with room for uncertainty. Zurich's homicide calculation is separate.

Making the choice personal

A school visit and a trial journey can help settle a close choice. Can you catch the bus after a club finishes? Are pool fees affordable? Check admission, language, friendships and wellbeing too.

Zurich stays first when we double the importance of schooling or travel separately. Three hypothetical 2035 scenarios keep Zurich, Madrid and Singapore at the top; Dublin and Paris switch places in one. These explore changes in services, costs, schools and air quality. Use the shortlist to plan a school day that works for your family.

References

[1] Team-supplied city-indicator table, sixteen cities. Input table, November 2026; supplied values, years and flags retained.
[2] T. L. Saaty. The analytic hierarchy process: what it is and how it is used. *Mathematical Modelling*, 9(3–5), 161–176, 1987. https://doi.org/10.1016/0270-0255(87)90473-8.
[3] C.-L. Hwang and K. Yoon. *Multiple Attribute Decision Making: Methods and Applications*. Springer, 1981. https://doi.org/10.1007/978-3-642-48318-9.
[4] Entur. National Stop Place Registry, Oslo export dated 9 November 2026; NeTEx stop-place and parent-group definitions. https://developer.entur.org/pages-nsr-nsr/; https://enturas.atlassian.net/wiki/plugins/viewsource/viewpagesrc.action?pageId=3846668311.
[5] Czech Statistical Office. *Schools and school facilities, 2024/25*, Tables 3.1.3 and 6.1.18. https://csu.gov.cz/docs/107508/299702e7-bc0a-1dc2-87db-600d0303f2f4/23004225.pdf?version=1.1.
[6] Singapore Department of Statistics. *Census of Population 2020, Statistical Release 2*, Tables 93 and 124; official Parliament mirror and official data API. https://www.parliament.gov.sg/docs/default-source/default-document-library/cop2020sr2.pdf.
[7] OpenAI. Codex, AI-assisted research, coding, checking and writing environment. Used November 2026. https://openai.com/codex/.
[8] Exa. `web_fetch_exa`, official-source retrieval and text extraction. Used November 2026. https://exa.ai/.
[9] PBS/TRAKO for the Warsaw Metropolitan Area. *Raport nr 2 z diagnozy zachowań transportowych mieszkańców*, July 2022, pp. 3–4, 58, 64. Fieldwork 12 March–30 April 2022; city school-travel sample base 22. Original retrieved 9 November 2026. https://omw.um.warszawa.pl/wp-content/uploads/2022/09/Raport-nr-2-glowny.pdf
[10] Statistics Netherlands (CBS). *Meer mensen vermoord in 2022, meeste slachtoffers in Rotterdam*, 29 August 2023; provisional annual city table, and accompanying *Slachtoffers moord en doodslag 2022*, Table 7, period 2018–2022. Originals retrieved 9 November 2026. https://www.cbs.nl/item?sc_itemid=209e5ed2-506c-42d6-b414-49d3556810bf https://www.cbs.nl/nl-nl/maatwerk/2023/35/slachtoffers-moord-en-doodslag-2022
[11] Kantonspolizei Zürich. *Polizeiliche Kriminalstatistik Kanton Zürich 2025*, pp. 117–118, city totals and definition of a criminal act. Original retrieved 9 November 2026. https://www.zh.ch/content/dam/zhweb/bilder-dokumente/themen/sicherheit-justiz/delikte-praevention/dokumente/pks/aktuell/jahresbericht_2025.pdf
[12] Eurostat. *Transport—cities and greater cities*, dataset urb_ctran, TT1019V, average time of journey to work (minutes). Retrieved 10 November 2026: IE001C/2011, 28.8; CH002C/2024, 39.9, flag e (estimated); NO001C/2009, 31.0. Current labels are Dublin, Zurich and Oslo (greater city); exact historical boundary and survey window are not established here. https://ec.europa.eu/eurostat/databrowser/view/urb_ctran/default/table?lang=en.
[13] Tokyo Metropolitan Government, Bureau of Transportation. *Oedo Line station list, Toei Subway ticket-service directory*. Original CSV, 38 named stations; catalogue created 1 September and updated 4 September 2026; inventory reference date unstated. Retrieved 9 November 2026 UTC. CC BY (version unspecified). https://spec.api.metro.tokyo.lg.jp/spec/t000018d0000000027-2911e010ab42b6f8fad459fc7dc10e83-0?lang=en; https://www.opendata.metro.tokyo.lg.jp/kotsu/subway/subway_branch_o_stationlist.csv.
[14] Metro Warszawskie. *M1 and M2 route station directories*; retrieved 9 November 2026 UTC. https://metro.waw.pl/metro-warszawskie/linia-m1/przebieg/; https://metro.waw.pl/metro-warszawskie/linia-m2/przebieg/. *Network plan*, marked April 2024; the hosting path is not its content date. https://metro.waw.pl/wp-content/uploads/2026/02/PLAN-METRA_2.pdf.
[15] M. Bruno, H. P. Monteiro Melo, B. Campanelli and V. Loreto. A universal framework for inclusive 15-minute cities. *Nature Cities*, 1, 633–641, 2024. https://doi.org/10.1038/s44284-024-00119-4. Author-platform resident grids for Singapore and Zurich, accessed 7 November 2026: https://whatif.sonycsl.it/15mincity/.
[16] A. Bakken. *Ung i Oslo 2023: Ungdomsskolen og videregående*, NOVA Rapport 6/23, OsloMet, pp. 1, 27, 33, 79–80. Vg3 is the final upper-secondary year. Survey participants: 14,667 lower-secondary (78% response) and 10,595 upper-secondary (60% response). https://oda.oslomet.no/oda-xmlui/bitstream/handle/11250/3065089/NOVA-Rapport-6-2023.pdf.
[17] S. Opricovic and G.-H. Tzeng. Compromise solution by MCDM methods: A comparative analysis of VIKOR and TOPSIS. *European Journal of Operational Research*, 156(2), 445–455, 2004. https://doi.org/10.1016/S0377-2217(03)00020-1.
[18] City of Oslo, municipal transport GIS, layer 800; Norwegian Mapping Authority, municipality 0301 boundary. Accessed November 2026. https://geodata.bymoslo.no/arcgis/rest/services/geodata/Temadata_Publikum/MapServer/800.
[19] Verkehrsbetriebe Zürich. *Zahlen und Fakten 2025*, p. 6. https://www.stadt-zuerich.ch/content/dam/vbz/de/die-vbz/zahlen-fakten/geschaeftsbericht/Z4690_VBZ_FB-2025_Web.pdf.
[20] Metropolitan Area Transport Ltd. *Metrolla*. Official operator inventory, accessed November 2026. https://kaupunkiliikenne.fi/liikennointi/metrolla/.
[21] City of Stockholm. *Simhallarna* and current swimming overview. Accessed 8 November 2026. https://motionera.stockholm/trana-gymma-simma/simhallarna/.
[22] Atelier parisien d'urbanisme. *Atlas des activités à Paris*, March 2022, p. 37. Sources RATP 2021 and Apur 2022. https://www.apur.org/sites/default/files/atlas_activites_paris_immobilier.pdf.
[23] RKA/Kolada, indicator N17816, Stockholm municipality 0180, period 2025; source Skolverket (Siris). Definition and total-provider scope archived with the API response. https://api.kolada.se/v3/data/kpi/N17816/municipality/0180/year/2025.
[24] Académie de Paris. *Chiffres clés 2025*, 2024 school-entry pupils and 31 December 2024 teacher totals. https://www.ac-paris.fr/sites/ac_paris/files/2025-08/chiffres-cl-s-acad-mie-de-paris-44994.pdf.
[25] Dublin City Council and National Transport Authority. *Dublin City Centre Transport Plan 2023, Technical Note 6*, issued 2024, Figure 3-7. https://www.dublincity.ie/sites/default/files/2024-04/06-dublin-city-centre-transport-plan-public-transport-technical-note.pdf.
[26] Shanghai Municipal Education Commission. Regular secondary-school pupil and teacher tables, school year 2025. https://edu.sh.gov.cn/xxgk2_zhzw_tjsj_01/20260610/890926a33cbf443d9abf07c6f65b7734.html.
[27] Singapore Ministry of Education. Secondary enrolment by level and age, 2020. https://data.gov.sg/datasets/d_cc84fc751ef5106d9e26cfdbf61776c8/view.

[28] Istanbul Metropolitan Municipality. 2012 household transport survey, official data-portal response resource a75118db-28c9-4fdf-b367-d5933b7bb810; questionnaire, report and nonidentifying aggregate audit archived.
[29] Consorcio Regional de Transportes de Madrid. M4 Estaciones, register dated 29 May 2026; municipality code and network-station identifiers. Accessed 9 November 2026. https://services5.arcgis.com/UxADft6QPcvFyDU1/arcgis/rest/services/M4_Red/FeatureServer/0.
[30] Instituto Nacional de Estadística. Transport statistics metadata, municipal classification, 2026. https://www.ine.es/dynt3/metadatos/RespuestaDatos.html?oper=297.
[31] Nerima City. Municipal swimming inventory and linked venue pages, accessed 9 November 2026. https://www.city.nerima.tokyo.jp/shisetsu/koen/index.html.
[32] Nerima City. Heiwadai closure, 27 October 2026; Miharadai planned closure, 29 September 2026. https://www.city.nerima.tokyo.jp/kankomoyoshi/sports/oshirase/20261027.html; https://www.city.nerima.tokyo.jp/kankomoyoshi/sports/oshirase/20260929.html.
[33] Department of Education and Youth, Ireland. *Teacher Workforce Data Quality Report*, 2026, Sections 5.2 and 8.4. https://nationaldata.ie/media/NationalDataie/MethodologyDocumentation/Teacher_Workforce_Data_Quality_Report.pdf.
[34] Wiener Linien. Versioned operator OGD files and *Beschreibung der OGD-Dateien*, v1.3 (2022), p. 4. Export version 8 November 2026; accessed 9 November 2026. DIVA areas group stopping positions. GTFS and passenger timetable catalogue retrieved 11 November 2026; GTFS service-day subset dated 10 November. https://www.wienerlinien.at/ogd_realtime/doku/; https://www.wienerlinien.at/fahrplaene.
[35] City of Vienna. Existing U-Bahn stops and Vienna boundary, OGD WFS layers UBAHNHALTOGD and LANDESGRENZEOGD. Accessed 9 November 2026; boundary record dated 1 September 2026. Datenquelle: Stadt Wien – data.wien.gv.at. https://data.wien.gv.at/daten/geo?service=WFS&request=GetCapabilities.
[36] Prague Integrated Transport (PID). *Zastávky v Praze*, Prague-only station lists (9 November 2026), native stop data and GTFS service dated 10 November 2026; public request-stop policy from June 2024. https://pid.cz/o-systemu/opendata/; https://pid.cz/vsechny-zastavky-autobusu-a-trolejbusu-v-ramci-pid-budou-na-znameni/.
[37] Dopravní podnik hlavního města Prahy. *Annual Report 2025*, PDF pp. 17 and 103. Line appearances and distinct interchange complexes retained separately. https://www.dpp.cz/spolecnost/o-spolecnosti/vyrocni-zpravy.
[38] Dopravní podnik hlavního města Prahy. Flora station closure from 2 February 2026 until further notice; Florenc C partial accessibility restriction. Accessed 9 November 2026. https://www.dpp.cz/omezeni-a-mimoradne-udalosti/detail/15; https://www.dpp.cz/omezeni-a-mimoradne-udalosti/detail/39.
[39] Berliner Verkehrsbetriebe (BVG). U1–U9 official route profiles and tariff-zone definitions. Native station identifiers, station-level zone A and interchange declarations; accessed 9 November 2026. https://www.bvg.de/de/verbindungen/linienuebersicht https://www.bvg.de/de/abos-und-tickets/tarifzonen-und-tarifbestimmungen
[40] Ministère de l'Éducation nationale, DEPP. Institution personnel indicators and general/technological lycée pupil counts, rentrée 2025; Henri-IV UAI 0750654D. Teacher ETP covers the institution, not a published upper-stage allocation. Accessed 9 November 2026. https://data.education.gouv.fr/explore/dataset/fr-en-indicateurs_personnels_etablissements2d/ https://data.education.gouv.fr/explore/dataset/fr-en-lycee_gt-effectifs-niveau-sexe-lv/
[41] Ministère de l'Éducation nationale. Official school directory, Henri-IV UAI 0750654D, row updated 7 November 2026; post-baccalaureate flag. Accessed 9 November 2026. https://data.education.gouv.fr/explore/dataset/fr-en-annuaire-education/
[42] Transport for London. Unified API, tube-mode StopPoint inventory and StopTypes metadata; accessed 9 November 2026. Native station and hub identifiers distinguish station entities from platforms and entrances. https://api.tfl.gov.uk/StopPoint/Mode/tube https://api.tfl.gov.uk/StopPoint/Meta/StopTypes
[43] Office for National Statistics. Regions (December 2025) Boundaries EN BFC, full-resolution coastline-clipped geometry; London region E12000007. Accessed 9 November 2026. https://services1.arcgis.com/ESMARspQHYMw9BZ9/arcgis/rest/services/Regions_December_2025_Boundaries_EN_BFC/FeatureServer/0
[44] Land Transport Authority, Singapore. Bus Stop Location, Data.gov.sg catalogue and GeoJSON; retrieved 7 November 2026. Archived original geometry and catalogue HTML; count uses distinct nonempty native bus-stop codes. https://data.gov.sg/datasets/d_3f172c6feb3f4f92a2f47d93eed2908a/view
[45] Sport Singapore, ActiveSG Circle. Swimming pool facility directory and 25 linked venue details, retrieved 10 November 2026. https://www.activesgcircle.gov.sg/facilities/swimming-pools.
[46] The Kallang. OCBC Aquatic Centre and swimming information, captured 10 November 2026; inventory date unstated. https://www.thekallang.com.sg/en/plan-your-visit/venues-and-amenities/ocbc-aquatic-centre.html; https://www.thekallang.com.sg/en/things-to-do/sports/swimming.html.
[47] Sport Singapore, ActiveSG Circle. *Editors' pick: must-hit pools in Singapore*, published 23 April 2025; retrieved 8 November 2026. Archived provider-extracted text and linked venue details; current pages corroborate venue identity, not 2025 opening hours. https://www.activesgcircle.gov.sg/read/editors-pick-must-hit-pools-in-singapore
[48] Statistical Office in Warszawa. *Statistical Yearbook of Warsaw 2025*, education workbook and PDF pp. 156–160, 165; rounding note p. 22. School-year 2024/25 tables; archived original workbook. https://new.stat.gov.pl/en/publications-portal/statistical-yearbook-warsaw-2025
[49] City of Zürich. Municipal indoor and school-pool directory and linked venue details. Original HTML retrieved 10 November 2026, including Wärmebad Käferberg; public-session and holiday restrictions retained. https://www.stadt-zuerich.ch/de/stadtleben/sport-und-erholung/sport-und-badeanlagen/hallenbaeder.html
[50] Milanosport. Swimming-pool tariffs effective 1 June 2025, updated 9 September 2025. Retrieved 8 November 2026; archived provider-extracted PDF representation, tariff groups A and B. https://www.milanosport.it/workspace/uploads/nuovi-tariffari-1-giugno-2025-agg-9-10-2025.pdf
[51] City of Stockholm and Local Police Area Vällingby. *Lägesbild 2025*, PDF p. 11 (printed p. 10), Bromma and Hässelby-Vällingby metro stations; retrieved 8 November 2026. Archived original report. https://meetingspublic.stockholm.se/welcome-sv/namnder-styrelser/hasselby-vallingby-stadsdelsnamnd/mote-2025-12-18/agenda/6-lagesbild-2025-samverkansoverenskommelse-mellan-hasselby-vallingby-sdf-och-lokalpolisomrade-vallingbypdf?downloadMode=open

[52] Comune di Milano. Stazioni_Metro, ATM_Metro MapServer layer 0; metadata date 16 October 2024. Original station and ID responses retrieved November 2026; municipal code 015146 and five operator-map interchange pairs retained. https://gisportal.comune.milano.it/server/rest/services/D_Mobilita/ATM_Metro/MapServer/0. ATM, metro/suburban network map, V26B label, publication date unreported. https://www.atm.it/it/AltriServizi/Trasporto/Documents/schema_rete_metro.pdf.
[53] Department of Education and Youth. *The Teaching Workforce in Ireland: Primary and Post Primary Level, 2025/26 School Year*. April 2026, Table 10, p. 14; November 2025 payroll. https://assets.gov.ie/static/documents/5935f213/Teacher_Workforce_Data_Report_2026.pdf.
[54] Statistics Finland / Eurostat. Finland UOE national reference metadata, Annex III–Additional tables, sheet 15.2.2, cell F10. National metadata updated 1 October 2026; original workbook retrieved 9 November 2026. Full-time/part-time and FTE estimation methods. https://ec.europa.eu/eurostat/cache/metadata/Annexes/educ_uoe_enr_esmsed_fi_an_Annex_III_-_Additional_tables.xlsx.
[55] Gemeente Amsterdam. *Tram en metro: haltes*, Maps Amsterdam open geodata, TRAMMETRO_PUNTEN_2026; metadata ID 649, 2026, exact observation date unreported; municipal boundary INDELING_GEMEENTE, March 2026. Original GeoJSON and metadata retrieved 9 November 2026. https://maps.amsterdam.nl/open_geodata/?k=649.
[56] Statistics Netherlands (CBS) / PDOK. *CBS Gebiedsindelingen 2025*, non-generalised municipal boundary, GM0363 Amsterdam; retrieved 9 November 2026. CC BY 4.0. https://service.pdok.nl/cbs/gebiedsindelingen/2025/wfs/v1_0?service=WFS&request=GetCapabilities.
[57] Mazovian Provincial Headquarters, State Fire Service. *Information bulletin for 2025*, published February 2026, physical p. 4. Locates 21 M1 and 18 M2 station appearances within Warsaw; the shared interchange is consolidated using the operator directories. https://www.gov.pl/attachment/2b7775c0-aa52-487e-809a-66df1b991b59.
[58] Numbeo. *Property Prices Index by City*, annual 2026 and rolling Current. Archived copies; access date recorded as 7 November 2026. https://www.numbeo.com/property-investment/rankings.jsp?title=2026; https://www.numbeo.com/property-investment/rankings_current.jsp.

Report on Use of AI Tools

The team used AI-assisted retrieval, calculation and drafting during 8–14 November 2026. We checked outputs against official source records, saved retrieval receipts, recomputed model results, and compiled and reviewed the rendered manuscript. The tools below record the uses documented in the team report.[7, 8]

Table 9: Recorded AI-tool use and checks.

Tool	Who/when (as recorded)	Used for	How we checked it
OpenAI Codex	Team record, 8–11 and 14 November 2026	Official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, wording revisions, and simulated judge reviews of the draft	Compared claims with source records; recomputed calculations; compiled and reviewed the rendered pages
Anthropic Claude (Claude Code)	14 November 2026	Planning the revision after our advisor's feedback, AHP and scenario calculations, all figures, proxy-data searches, and small text edits	Recomputed every reported number from the result files; inspected each figure at print size
Exa (`web_fetch_exa`)	Team record, 8–9 November 2026	Retrieval and text extraction from official webpages, PDFs and data interfaces	Saved source URLs and returned text; checked extracted representations against original downloaded records and excluded ambiguous material
Anti-defensive-writing skill	Applied during manuscript revision	Editorial guidance for concise, evidence-supported wording	Reviewed claims against the manuscript evidence and retained conclusion-changing checks

Usage by other team members: to be added by the team before submission.